

Mathematics A

Unit 2 • Chapter OL-T04 Classified

Cross-session • chapter-organised candidate

26 classified items • 53 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 2 • Unit 2 • Chapter OL-T04 • 26 items • 53 marks

Chapter	Items	Marks	Starts
01. OL-T04 Powers, Roots & Standard Form	26	53	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Powers, Roots & Standard Form

Unit 2 • 53 marks • 26 item(s)

June 2025 | 4MA1-1H Q8 | 2 marks

Ordinary number to standard form

June 2025 | 4MA1-1H Q8 | 2 marks

Multiply or divide standard-form numbers

June 2024 | 4MA1-1H Q9 | 3 marks

Standard form to ordinary number

November 2023 | 4MA1-1H Q9 | 1 marks

Standard form to ordinary number

November 2023 | 4MA1-1H Q9 | 1 marks

Ordinary number to standard form

November 2023 | 4MA1-1H Q9 | 2 marks

Multi-step contextual standard-form problem

June 2023 | 4MA1-1H Q15 | 3 marks

Multiply or divide standard-form numbers

June 2023 | 4MA1-2H Q6 | 2 marks

Ordinary number to standard form

November 2024 | 4MA1-1H Q9 | 1 marks

Standard form to ordinary number

November 2024 | 4MA1-1H Q9 | 2 marks

Multiply or divide standard-form numbers

June 2022 | 4MA1-1H Q8 | 1 marks

Ordinary number to standard form

June 2022 | 4MA1-1H Q8 | 1 marks

Standard form to ordinary number

January 2022 | 4MA1-1H Q8 | 2 marks

Multiply or divide standard-form numbers

January 2022 | 4MA1-2H Q12 | 3 marks

Raise a standard-form quantity to a power

January 2020 | 4MA1-1H Q6 | 1 marks

Standard form to ordinary number

... 11 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

8 (a) Write 520 million in standard form.

.....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

.....
(1)

WORKING SPACE

- (c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....
(2)

WORKING SPACE

Question 05

4MA1-1H Q9

MODULAR UNIT 2

3 marks

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes
(2)

WORKING SPACE

Question 06

4MA1-1H Q9 (a)

MODULAR UNIT 2

1 marks

- 9 (a) Write 5.87×10^{-4} as an ordinary number.

.....
(1)

WORKING SPACE

Question 07

4MA1-1H Q9 (b)

MODULAR UNIT 2

1 marks

(b) Write 84 000 000 in standard form.

.....
(1)

WORKING SPACE

Question 08

4MA1-1H Q9 (c)

MODULAR UNIT 2

2 marks

The number of neurons in a human brain is 8.5×10^{10}
The number of neurons in a monkey brain is 1.47×10^9

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

- (c) Work out the value of K
Give your answer correct to one decimal place.

$K = \dots\dots\dots$
(2)

WORKING SPACE

Question 13

4MA1-1H Q15

MODULAR UNIT 2

3 marks

- (b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

(3)

WORKING SPACE

Question 20

4MA1-2H Q6

MODULAR UNIT 2

2 marks

6 (a) Write 76 000 000 in standard form.

.....
(1)

(b) Write 5.4×10^{-4} as an ordinary number.

.....
(1)

WORKING SPACE

Question 08

4MA1-1H Q9 M01

MODULAR UNIT 2

1 marks

- 9 (a) Write 8.4×10^{-5} as an ordinary number.

.....
(1)

WORKING SPACE

Question 09

4MA1-1H Q9 M02

MODULAR UNIT 2

2 marks

- (b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

.....
(2)

WORKING SPACE

Question 09

4MA1-1H Q8 Q8(a)

MODULAR UNIT 2

1 marks

8 (a) Write 0.000089 in standard form.

.....

(1)

WORKING SPACE

Question 10

4MA1-1H Q8 Q8(b)

MODULAR UNIT 2

1 marks

(b) Write 8.34×10^4 as an ordinary number.

.....
(1)

WORKING SPACE

Question 07

4MA1-1H Q8

MODULAR UNIT 2

2 marks

8

$$a = 4.2 \times 10^{-24}$$

$$b = 3 \times 10^{145}$$

Work out the value of $a \times b$

Give your answer in standard form.

WORKING SPACE

Question 26

4MA1-2H Q12

MODULAR UNIT 2

3 marks

12 $a = 6 \times 10^{40}$

Work out the value of a^3

Give your answer in standard form.

WORKING SPACE

6 (a) Write 7.8×10^{-4} as an ordinary number.

(1)

YOUR WORKING

(b) Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$

Give your answer in standard form.

(2)

YOUR WORKING

6 (a) Write 2 840 000 000 in standard form.

(1)

YOUR WORKING

(b) Write 2.5×10^{-4} as an ordinary number.

(1)

YOUR WORKING

(b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

(3)

YOUR WORKING

8 The table shows the populations of five countries.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

- (a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

(2)

YOUR WORKING

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

(b) work out the value of k .

Give your answer correct to the nearest whole number.

$k = \dots\dots\dots$
(2)

YOUR WORKING

- 9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules
(2)

YOUR WORKING

- 9 A rainwater tank contains 2.4×10^7 raindrops.
The rainwater tank also contains 1.75×10^6 bacteria.

- (a) Work out the number of bacteria per raindrop in the tank.
Give your answer in standard form correct to 2 significant figures.

.....
(3)

A drop of rainwater contains 5.01×10^{21} atoms.

In a drop of rainwater the number of atoms is 3 times the number of molecules.

- (b) Work out the number of molecules in the rainwater tank.
Give your answer in standard form correct to one significant figure.

..... molecules
(2)

YOUR WORKING

5 (a) Write 5.7×10^{-3} as an ordinary number.

.....
(1)

(b) Write 800 000 in standard form.

.....
(1)

(c) Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

.....
(2)

.....
(Total for Question 5 is 4 marks)

YOUR WORKING

9 (a) Write 6.25×10^{-4} as an ordinary number.

.....
(1)

(b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^4)$
Give your answer in standard form.

.....
(2)

(Total for Question 9 is 3 marks)

YOUR WORKING

8 (a) Write 5.76×10^4 as an ordinary number.

.....
(1)

(b) Work out $\frac{3 \times 10^5 + 8 \times 10^3}{4 \times 10^{-2}}$

Give your answer in standard form.

.....
(2)

(Total for Question 8 is 3 marks)

YOUR WORKING
